Electrically Heated Ethanol Dehydration for Sustainable Bioethylene Production

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1. Introduction

The substantial reliance on fossil fuels to meet global energy demands hinders climate neutrality targets [1]. A profound shift is needed, involving renewable energy adoption, advanced heating technologies, and regenerative feedstocks. Electrified catalytic reactors, utilizing resistive, inductive, and microwave heating, offer a promising way forward. While extensive research has focused on resistive heating in electrically conductive wall reactors and structured catalysts [2-3], limited attention has been given to resistively heated catalytic fixed-bed reactors [4]. Implementing resistive heating in catalyst beds made of granules is technically challenging due to the need for physical continuity between conductive particles to avoid hot spots [5]. Our goal is to electrify the endothermal bioethanol-to-bioethylene reaction using an electrically heated catalytic fixed-bed for efficient, productive, and CO_2-free operation with alumina catalyst.

2. Methods

A tubular reactor was used to evaluate the bed resistivity of a physical mixture along the axial length, varying volume fractions and particle sizes to determine the optimal parameters for a conductive and reproducible bed. The catalyst bed consisted of a homogeneous mixture of electrically conductive graphite particles and commercial alumina catalyst. Catalytic investigations with resistive heating were carried out in a nearly adiabatic double-tube reactor with radial heat insulation, designed for a 5 ml fixed-bed. Further studies scaled up the reactor capacity from 5 ml to 500 ml.

3. Results and discussion

Resistivity measurements revealed that a volume fraction of 30% conductive particles is crucial for achieving the desired bed resistivity. The successful use of resistive heating in an electrically conductive catalyst bed for endothermal ethanol dehydration

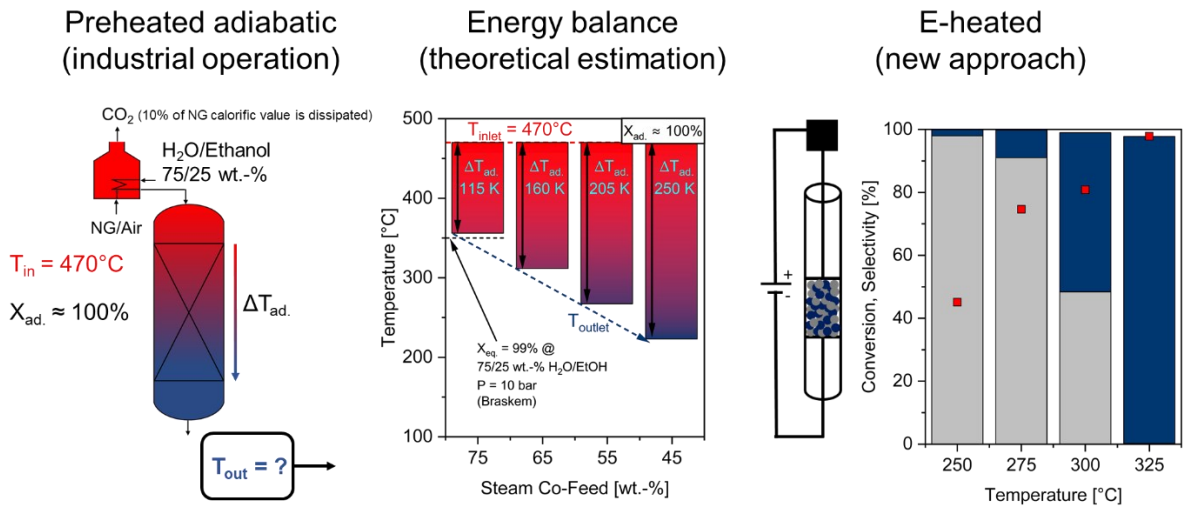

Figure 1: Comparison of the conventional adiabatic process and resistive heating on ethanol dehydration to ethylene.

demonstrated nearly complete conversion to ethylene at 325°C, with a homogeneous temperature distribution and no detectable hotspots. Unlike the industrial adiabatic process [6], which relies on 75 wt.-% steam as a heat carrier to prevent temperature gradients, the resistive heating approach utilized pure ethanol. Additionally, 10% of natural gas (NG) calorific value stored in the flue gas, which is typically dissipated into the environment during preheating of the inlet stream, can be saved. In the conventional adiabatic process, performance depends on heat availability, whereas resistive heating is kinetically controlled, maximizing catalyst utilization. Resistive heating also allowed operation at a higher weight hourly space velocity (WHSV) while maintaining similar gas retention times, eliminating the need for steam and reducing energy input. The scale-up from 5 ml to 500 ml was successful, with axial temperature profiles recorded. Overall, resistive heating enables emission-free operation, ultimately intensifying the process and decarbonizing endothermal reaction heating.

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